

Topic 13: Variation of Parameters & Cauchy-Euler Equidimensional Equations

Professor Leonard Master Series

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: When Undetermined Coefficients Fails

Professor Leonard reveals why Variation of Parameters is the universal hammer:

"What do you do if $g(x) = \tan x, \sec x, \frac{1}{x}$, or $\ln x$? Derivatives of $\tan x$ never terminate—they blow up into $\sec^2 x, 2 \sec^2 x \tan x$, and you can never make a finite guess! Variation of Parameters NEVER guesses. It replaces constants c_1, c_2 with unknown functions $u_1(x), u_2(x)$ and integrates directly using the Wronskian!"

$$y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$$

$$u_1'(x) = -\frac{y_2(x)g(x)}{W(y_1, y_2)}, \quad u_2'(x) = \frac{y_1(x)g(x)}{W(y_1, y_2)}$$

(CRITICAL: The equation MUST be in standard form $y'' + P(x)y' + Q(x)y = g(x)$ with leading coefficient 1!)

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 13.1: Variation of Parameters with $\tan x$

Statement: Solve $y'' + y = \tan x$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Step-by-Step Whiteboard Solution:

- **Step 1: Complementary Solution:**

$$r^2 + 1 = 0 \implies y_1(x) = \cos x, \quad y_2(x) = \sin x$$

- **Step 2: Compute the Wronskian $W(y_1, y_2)$:**

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \cos^2 x - (-\sin^2 x) = \cos^2 x + \sin^2 x = 1$$

- **Step 3: Compute $u_1(x)$:**

$$u_1'(x) = -\frac{y_2 g}{W} = -\frac{\sin x \tan x}{1} = -\frac{\sin^2 x}{\cos x} = -\frac{1 - \cos^2 x}{\cos x} = -\sec x + \cos x$$

$$u_1(x) = \int (\cos x - \sec x) dx = \sin x - \ln |\sec x + \tan x|$$

- **Step 4: Compute $u_2(x)$:**

$$u_2'(x) = \frac{y_1 g}{W} = \frac{\cos x \tan x}{1} = \sin x$$

$$u_2(x) = \int \sin x dx = -\cos x$$

- **Step 5: Assemble $y_p = u_1 y_1 + u_2 y_2$:**

$$y_p(x) = (\sin x - \ln |\sec x + \tan x|) \cos x + (-\cos x) \sin x$$

$$y_p(x) = \sin x \cos x - \cos x \ln |\sec x + \tan x| - \sin x \cos x = -\cos x \ln |\sec x + \tan x|$$

- **Step 6: General Solution:**

$$y(x) = c_1 \cos x + c_2 \sin x - \cos x \ln |\sec x + \tan x|$$

Problem 13.2: Variation of Parameters with $\sec 2x$

Statement: Solve $y'' + 4y = \sec 2x$.

Step-by-Step Whiteboard Solution:

- **Step 1:** $y_1 = \cos 2x, y_2 = \sin 2x$.

$$W = \begin{vmatrix} \cos 2x & \sin 2x \\ -2 \sin 2x & 2 \cos 2x \end{vmatrix} = 2 \cos^2 2x + 2 \sin^2 2x = 2$$

- **Step 2: Compute $u_1(x)$:**

$$u_1' = -\frac{\sin 2x \sec 2x}{2} = -\frac{1}{2} \tan 2x \implies u_1(x) = \frac{1}{4} \ln |\cos 2x|$$

- **Step 3: Compute $u_2(x)$:**

$$u_2' = \frac{\cos 2x \sec 2x}{2} = \frac{1}{2} \implies u_2(x) = \frac{1}{2} x$$

- **Step 4: Final Solution:**

$$y(x) = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{4} (\cos 2x) \ln |\cos 2x| + \frac{1}{2} x \sin 2x$$

Problem 13.3: Cauchy-Euler Equidimensional Equation (Distinct Real Powers)

Statement: Solve $x^2 y'' - 2xy' - 4y = 0$ for $x > 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Recognize Cauchy-Euler Form:** Coefficients match the derivative order: $x^2 y''$, xy' , y .
- **Step 2: Trial Solution** $y = x^m$: $y' = mx^{m-1}$, $y'' = m(m-1)x^{m-2}$.

$$x^2[m(m-1)x^{m-2}] - 2x[mx^{m-1}] - 4x^m = 0$$

$$[m(m-1) - 2m - 4]x^m = 0$$

- **Step 3: Auxiliary Equation (Watch the $-m$!):**

$$m^2 - 3m - 4 = 0 \implies (m-4)(m+1) = 0 \implies m_1 = 4, \quad m_2 = -1$$

- **Step 4: General Solution:**

$$y(x) = c_1 x^4 + c_2 x^{-1} = c_1 x^4 + \frac{c_2}{x}$$

Problem 13.4: Cauchy-Euler Repeated Roots (The $\ln x$ Factor)

Statement: Solve $x^2 y'' + 5xy' + 4y = 0$ for $x > 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Auxiliary Equation:**

$$m(m-1) + 5m + 4 = 0 \implies m^2 + 4m + 4 = 0 \implies (m+2)^2 = 0 \implies m = -2, -2$$

- **Step 2: In Cauchy-Euler, repeated roots gain a factor of $\ln x$:**

$$y(x) = c_1 x^{-2} + c_2 x^{-2} \ln x = \frac{c_1 + c_2 \ln x}{x^2}$$

Problem 13.5: Cauchy-Euler Complex Conjugate Powers

Statement: Solve $x^2 y'' + xy' + 9y = 0$ for $x > 0$.

Step-by-Step Whiteboard Solution:

- **Step 1: Auxiliary Equation:**

$$m(m-1) + m + 9 = 0 \implies m^2 + 9 = 0 \implies m = \pm 3i$$

Real part $\alpha = 0$, Imaginary part $\beta = 3$.

- **Step 2: General Solution via $x^{i\beta} = e^{i\beta \ln x} = \cos(\beta \ln x) + i \sin(\beta \ln x)$:**

$$y(x) = c_1 \cos(3 \ln x) + c_2 \sin(3 \ln x)$$

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Forgetting to Divide by $a(x)$ in Variation of Parameters:** If $x^2y'' + \dots = g(x)$, you MUST divide by x^2 to find the true forcing function $f(x) = g(x)/x^2$ before using u'_1, u'_2 !
- **Trap 2: Cauchy-Euler $m(m-1)$ vs m^2 :** The auxiliary equation is $am(m-1) + bm + c = am^2 + (b-a)m + c = 0$. Students always forget to subtract a !